

Sub Code:	CS/IS/AI/AD/CI/CY31	Sub:	Laplace Transform and Vector Space	Section:	CSE & Allied Branches
Semester:	III	Term:	20.11.2023 to 02.03.2024	Marks:	30
Date:	26-02-2024	Time:	3:00 PM-4:00 PM	Test:	II

Q. No.		Questions	Blooms Level	CO's	Marks
1.	(a)	Define basis and dimension of a vector space V.	L1	CO3	2
	(b)	Find the Eigen vector corresponding to the Eigen value $\lambda = 3$ of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.	L2	CO5	3
	(c)	Determine the matrix that defines a reflection in the y-axis followed by rotation through an angle $\pi/3$ and then by the dilation of factor $7/2$. Find the image of the point $\begin{bmatrix} 5 \\ 2 \end{bmatrix}$ under this transformation.	L3	CO3	5
	(d)	Construct orthonormal basis for the following using Gram-Schmidt process $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$, $w = \text{span}\{x_1, x_2, x_3\} \in R^3$.	L4	CO4	5

2.	(a)	Write any two properties of Eigen values.	L1	CO5	2
	(b)	Prove that the transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(x, y) = (x - 2y, 4x)$ is linear.	L2	CO3	3
	(c)	Find the least-squares solution of $Ax = b$, where $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$.	L3	CO4	5
	(d)	Consider the bases $B = \{(1, 2), (3, -1)\}$ and $B' = \{(3, 1), (5, 2)\}$ of $\mathbb{R}^2$. Find the transition matrix from B to B' . If u is a vector such that $u_B = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, find $u_{B'}$.	L4	CO3	5

3.	(a)	Define orthogonal basis with an example.	L1	CO4	2
	(b)	Find the null space of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 2 & 4 & 8 & 12 \\ 3 & 6 & 7 & 13 \end{bmatrix}$.	L2	CO4	3
	(c)	Determine the kernel and range of each of the transformation $T(x, y) = (5x, x+y, 2y)$ of $R^2 \rightarrow R^3$ and hence verify Rank nullity theorem.	L3	CO3	5
	(d)	Diagonalize the matrix $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.	L5	CO5	5